



Rafael Villagra de la Fuente

Advanced Aerospace Engineering Student

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LinkedIn

Engineering Portfolio

EDUCATION

•Politecnico di Torino & Universidad Nacional de Córdoba

09.2025 – Present

Double Degree Program: Master's in Aerospace Engineering (Weighted GPA: 29.1/30)

Turin, IT | Córdoba, AR

- Academic focus: Aerostructures and Aerogas dynamics.
- Key courses: Aeroelasticity, Design of Aerospace Vehicles.

•Universidad Nacional de Córdoba

03.2021 – Present

Bachelor's in Aeronautical Engineering (Weighted GPA: 7.03 - Eq. 25.4/30)

Córdoba, AR

- Key courses: Fluid Mechanics, Structural Analysis I and II, Control Theory, Flight Mechanics.
- International Student Assistant & Student Tutor (2024 - 2025).

PROFESSIONAL EXPERIENCE

•FDC Diseños y Desarrollos

12.2024 – 08.2025

Engineering Assistant - Internship

Córdoba, AR

- Designed the roll cage of a Dakar T3 category UTV from scratch, ensuring FIA T3 regulatory compliance while optimizing geometry to minimize mass and maximize torsional rigidity.
- Conducted suspension kinematics analysis using OptimumK, minimizing camber and toe angle variation throughout full suspension travel and verifying physical clearance constraints.
- Performed preliminary structural analysis using ANSYS Mechanical to evaluate component safety factors.
- Analyzed pre-season telemetry data with MoTEC to produce internal reports on tire wear and behavior as a function of vehicle setup parameters.

RELEVANT ACADEMIC PROJECTS

•Conceptual Design of a Regional / Hybrid Aircraft

2025

Multi-disciplinary parametric design study

- Built a parametric configuration matrix varying geometrical parameters to assess combined effects on MTOW and aerodynamic performance, using MATLAB.
- Identified feasible compromise configurations through multi-disciplinary trade-off analysis

•Flight mechanics and performance of the Douglas DC-3

2024

Performance, static and dynamic stability analysis

- Computed performance envelopes and longitudinal stability derivatives for multiple flight conditions using Python.
- Characterized static stability margins and dynamic longitudinal modes (phugoid and short-period), generating trimmed flight data curves.

•Lift of a parabolic airfoil using the discrete vortex method

2024

Computational aerodynamics

- Implemented a discrete vortex panel solver in MATLAB to compute lift coefficient and velocity field over a parabolic airfoil at varying angles of attack.
- Validated numerical results against thin airfoil theory from the literature, assessing solution convergence with panel density.

TECHNICAL SKILLS & CERTIFICATIONS

•Engineering Software (CAD/CAE/CFD): Ansys, SolidWorks, OpenFOAM, MSC Nastran and Patran

•Programming & Tools: MATLAB, Python, LaTeX

•Languages: Spanish (Native), English (B2 - FCE 178/190), Italian (B1 - PLIDA 102/120)

•Certifications: ANSYS CFD (35h) | OpenFOAM (5h) | Public Speaking — UNC

REFERENCES

•Javier Ciabattari, Engineer — FDC Diseños y Desarrollos

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•Juan Pablo Saldia, Professor & Course Director, Flight Mechanics — UNC

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•Andrés Cimino, Assistant Professor, Fluid Mechanics — UNC

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